

Online Campus Info System

A Full Stack Web Application for College Information, Admissions & Counselling

Team Members:

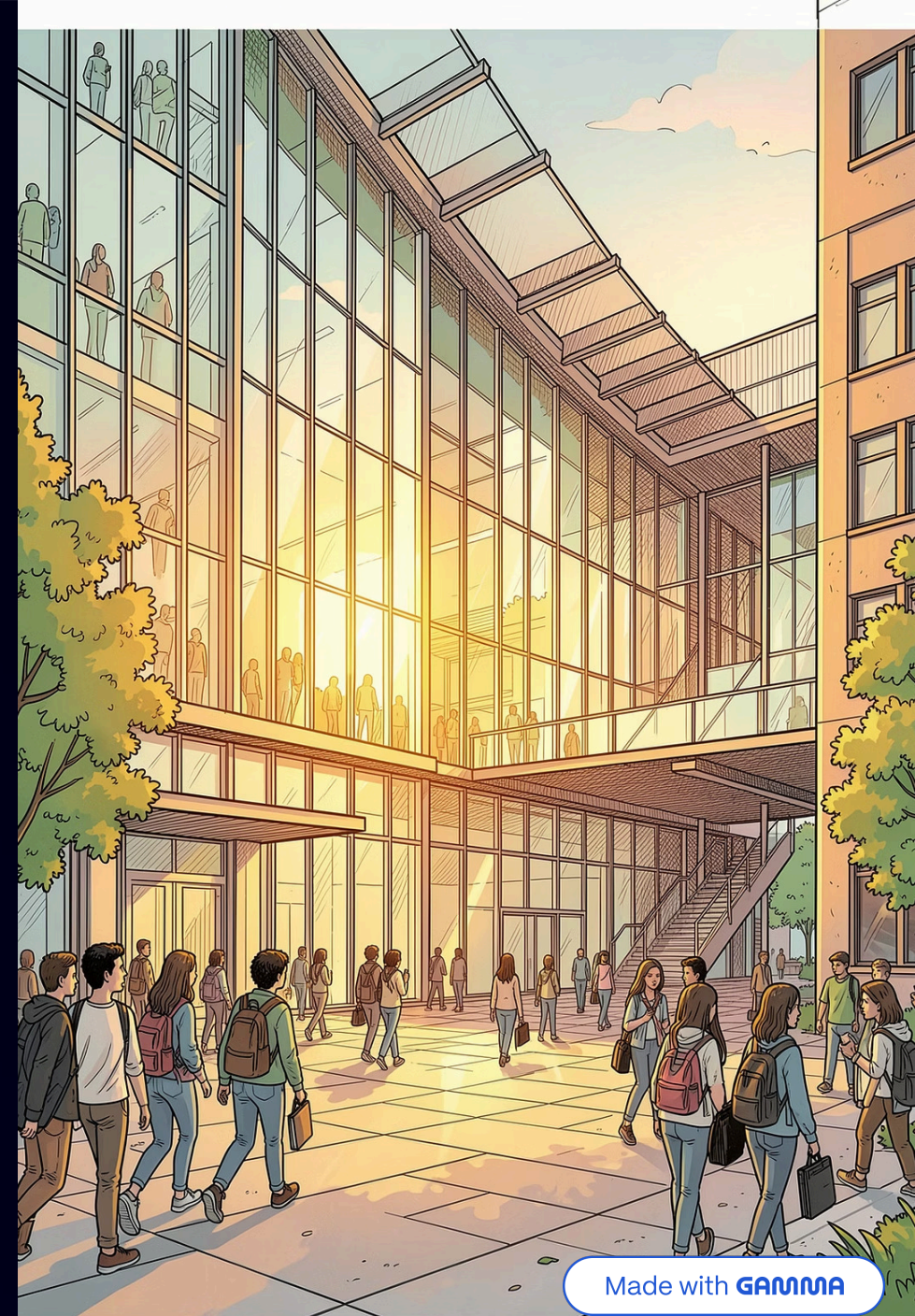
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Course: Full Stack Application Development (FSAD)

Semester: 2nd Semester



Problem Statement & Objective

The Problem

- Students lack a centralized platform to compare colleges, apply for admissions, and access counselling
- College authorities have no digital system to manage applications or showcase facilities

Our Objective

- Provide detailed college info – courses, facilities, and eligibility
- Enable digital admission applications
- Offer counselling support through query resolution
- Deliver feedback & comparison reports for informed decisions

Technology Stack

Frontend

React 19 + Vite 8 — Fast, modern SPA with hot module replacement

Backend

Spring Boot 3.4.1 + Spring MVC — Robust REST API layer

ORM & Database

Spring Data JPA + Hibernate 6.6 · MySQL 9.6

Security

Spring Security 6.x + JWT (jjwt 0.12.5) — Stateless, token-based auth

Build & Version Control

Maven (Backend) · npm (Frontend) · Git + GitHub

Key Dependencies

Backend — pom.xml

- **spring-boot-starter-web** — REST API
- **spring-boot-starter-data-jpa** — Database ORM
- **spring-boot-starter-security** — Auth & Authorization
- **spring-boot-starter-validation** — Input validation
- **mysql-connector-j** — MySQL driver
- **jjwt-api / impl / jackson** — JWT tokens
- **lombok** — Reduces boilerplate code

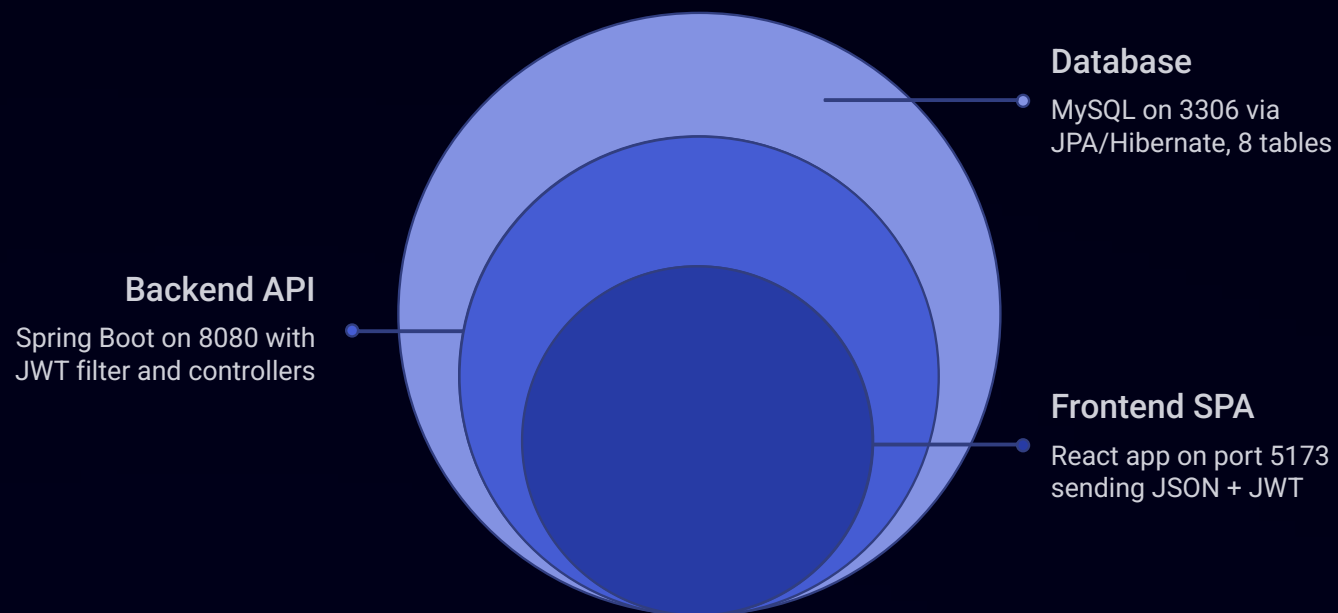
Frontend — package.json

- **react** — UI component library
- **react-router-dom** — Client-side routing
- **axios** — HTTP client for API calls
- **vite** — Build tool & dev server

 All dependencies are latest stable versions, ensuring security patches and performance improvements.

System Architecture

A clean **3-tier architecture** separates concerns across frontend, backend, and database layers, communicating via RESTful JSON APIs secured with JWT tokens.



The backend's layered structure — Security Filter → Controllers → Services → Repositories — ensures maintainability and testability across all 7 modules.

9 Modules Implemented

Registration & Login

JWT auth with 3 roles: Student,
Admin, Counsellor

College Data Updation

Admin CRUD for colleges, courses,
facilities & images

View College Details

Public browse, search, and detailed
college view

Student Application

Apply with course selection &
Statement of Purpose

Application Management

Admin reviews, accepts, or rejects
applications

Student Querying

Raise queries to counsellors

Counselling

Counsellors respond and mark
queries resolved

Feedback

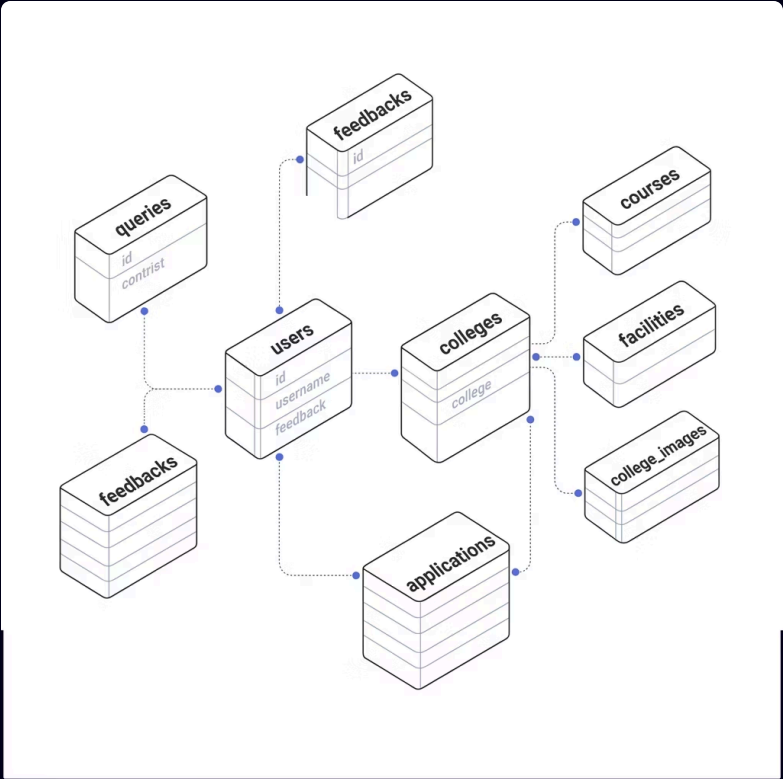
Rate colleges & counsellors (1–5
stars)

Reports

College comparison, counsellor
performance & statistics

Database Design

The schema comprises **8 normalized tables** with foreign key relationships, supporting all 9 modules and **40+ API endpoints** across 7 controllers.



users

id, name, email, password, role

college_images

id, image_url, caption (FK → colleges)

colleges

id, name, city, state, strength (FK → users)

applications

id, status, qualification, percentage (FK → users, colleges, courses)

courses

id, name, duration, fee, eligibility (FK → colleges)

queries

id, subject, message, response, status (FK → users)

facilities

id, type, name, capacity (FK → colleges)

feedbacks

id, type, rating, comment (FK → users, colleges)

User Roles & Workflow



Student

Register → Browse Colleges → View Details → Apply → Track Status → Raise Query → Get Response → Give Feedback → View Reports



Admin

Register → Add College → Add Courses/Facilities → Upload Images → View Applications → Accept / Reject



Counsellor

Register → View Assigned Queries → Respond → Track Performance

Key Features & Highlights



Security

- JWT-based stateless authentication
- Role-Based Access Control (RBAC)
- BCrypt password encryption
- CORS configured



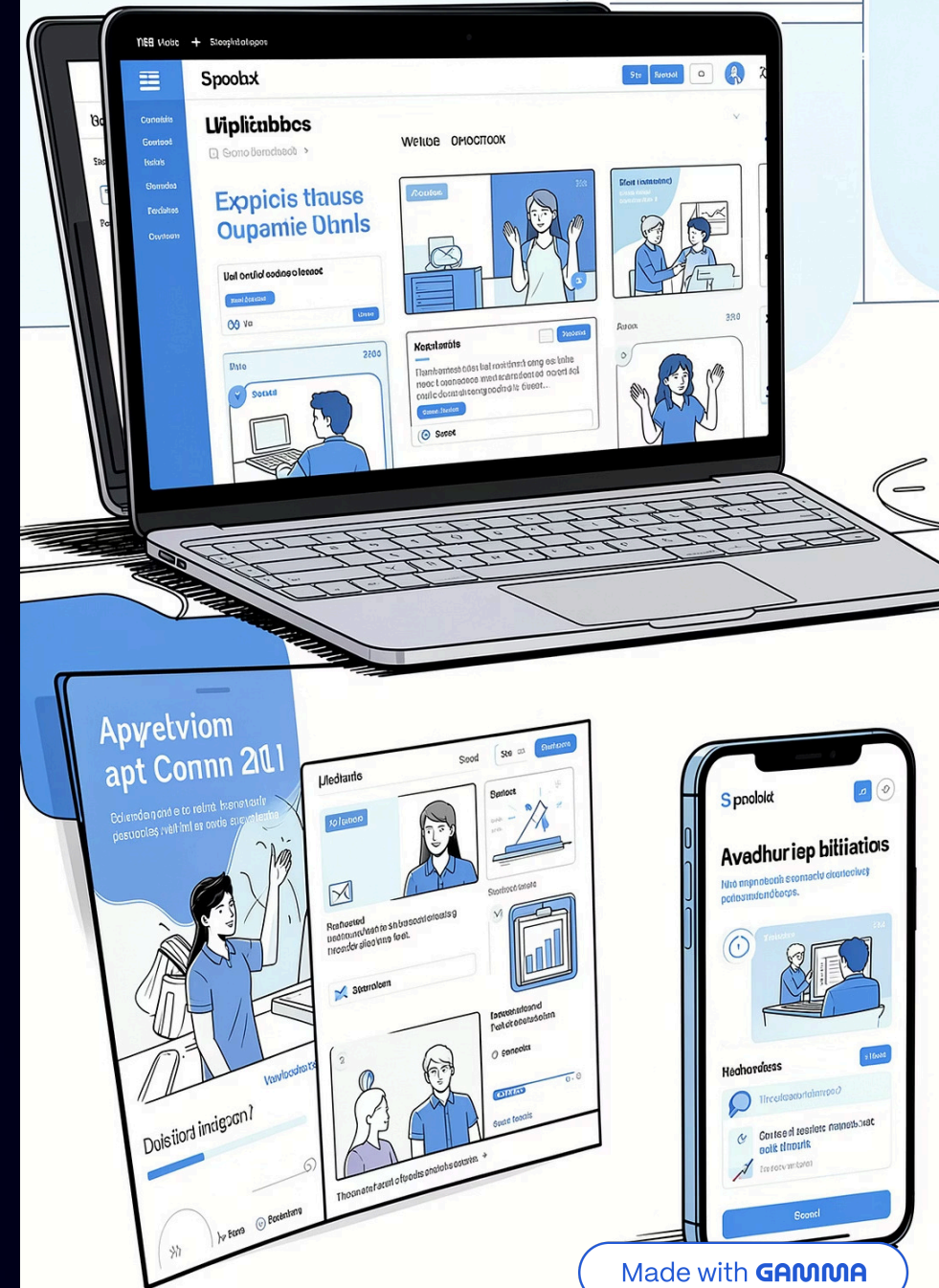
User Experience

- Responsive, mobile-friendly design
- Real-time form validation
- Color-coded status badges
- Star rating system & search/filter



Technical Excellence

- RESTful API design
- Global exception handling
- Auto-assign counsellors (least-load algorithm)
- File upload & Hibernate auto-DDL



Demo

How to Run

Terminal 1 — Backend:

```
cd backend  
mvn spring-boot:run
```

Terminal 2 — Frontend:

```
cd frontend  
npm run dev
```

Open **<http://localhost:5173>**

GitHub:

github.com/2025sl93041-gkb/Online-Campus-Info-System

Conclusion

The **Online Campus Info System** successfully implements all 9 required modules using a Spring Boot + React + MySQL stack — delivering a complete, production-ready solution for college information management, student admissions, counselling, and feedback analytics.

9

Modules

40+

API Endpoints

8

DB Tables

3

User Roles